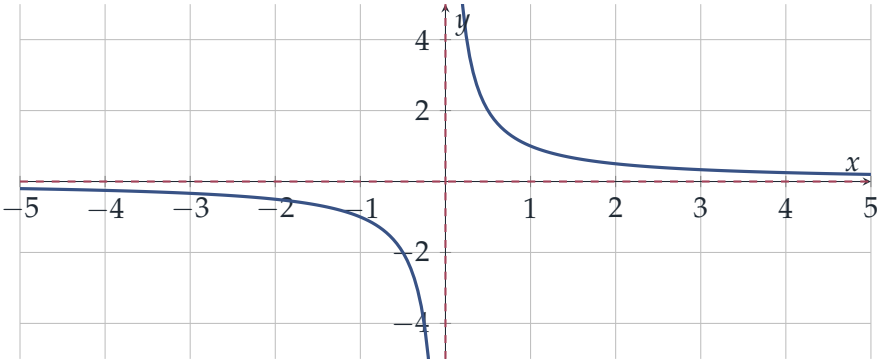
Chapter 4

Reciprocal function

Competences: determining a domain, drifting, studying a sign, drawing up a table of variations, representing

4.1 Reference course

The reciprocal function is defined by $f(x) = \frac{1}{x}$ on $\mathbb{R} \setminus \{0\}$.



Key ideas

$f'(x) = -\frac{1}{x^2} < 0$ for all $x \neq 0$. The reciprocal function is strictly decreasing on each intervals $]-\infty, 0[$ and $]0, +\infty[$. The rights $x = 0$ and $y = 0$ are asymptotes.

Frequent error

We cannot write that the reciprocal function is decreasing "on $\mathbb{R} \setminus \{0\}$ » without specifying the two intervals. $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$, which would contradict the overall definition of decline.

For a function $g(x) = P(x) + \frac{k}{x}$, we proceed in this order : domain ; derivative ; put to the same denominator and factorization; derivative sign; variations; boundary or boundary behavior; useful points; representation.

Worked example

On $]0, +\infty[$, $g(x) = x + \frac{4}{x}$ checks $g'(x) = 1 - \frac{4}{x^2} = \frac{(x-2)(x+2)}{x^2}$. The denominator is positive; g' is negative on $]0, 2[$ then positive on $]2, +\infty[$. The function allows a minimum $g(2) = 4$.

Session 23 — Definition and curve of the reciprocal function

Reciprocal function

15/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Determine the domain.

· Build the curve.

· Use symmetries.

Fluency · 6 to 8 minutes, without calculator

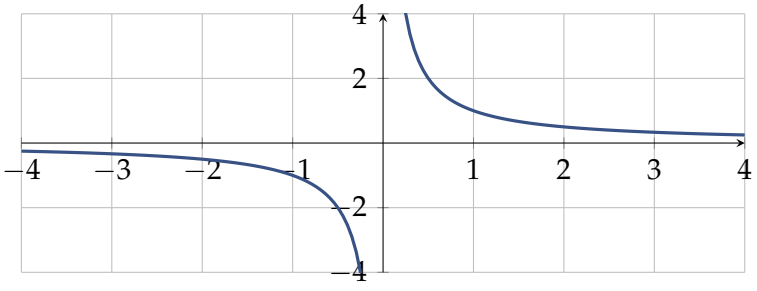
1. Reduce 3 $x + 5 - x + 2$. .....
2. To give the prohibited value of $\dfrac{x+1}{x}$. .....
3. Calculate $\dfrac{1}{5}$ and $-\dfrac{1}{5}$. .....
4. Derive 3 $x^2 - 5x + 1$. .....
5. Solve $(x-2)(x+1) = 0$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: determine the domain and build the curve.

Research activity

Calculate $\dfrac{1}{x}$ for $x =$ $-4, -2, -1, -\dfrac{1}{2}, \dfrac{1}{2}$, 1, 2, 4. Place points and describe symmetries.
$x =$ Is 0 absent?



Trace written

The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its curve is a symmetric hyperbole in relation to originally.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 5 + \frac{7}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{25}{x}$.

Reference in the manual

Supplements: Calao, courses p. 88·93; exercises p. 94, n° 1-5.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 101, exercise 42

The purpose of this exercise is to draw the curves representative of the functions f and g defined on $\mathbb{R}^*$ by: $f(x) = \frac{2}{x}$ and $g(x) = 4 - \frac{3}{x}$.

1. Complete the table of values for $x \in \{-5; -4; -2; -0{,}5; -0{,}2; -0{,}1; 0{,}1; 0{,}2; 0{,}5; 1; 2; 4; 5\}$.
2. Using a table of values, construct the curves representative of the functions f and g.

Personal work

Resuming Exercise 2 with the following data:
235 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 24 — Behavior at bounds and asymptotes

Reciprocal function

17/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Describe the behaviour at the bounds.

· Identify asymptotes.

· Read a representation.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 120 .....

2. Calculate $\frac{1}{6}$ and ~~$-\frac{1}{6}$~~. .....

3. Derive 3 $x^2 - 5x + 1$. .....

4. Solve $(x-2)(x+1) = 0$. .....

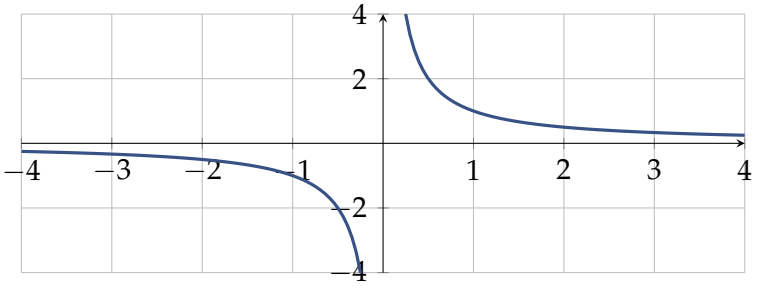
5. To give the prohibited value of $\frac{x+1}{x}$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reformulated: describe the behaviour at the bounds and identify the Asymptotes.

Research activity

Using a table of values close to 0 and for large values of $|x|$, describe $\frac{1}{x}$. Translate each behavior by an asymptote and a sentence.



Trace written

For $f(x) = \frac{1}{x}$, $x = 0$ is vertical asymptote and $y = 0$ horizontal asymptote. approach without cutting them off.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure

in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 6 + \frac{8}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{36}{x}$.

Reference in the manual

Supplements: Calao, courses p. 88·93; exercises p. 94, n° 4 to 9.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 95, exercise 16

We want to plot the representative curve of the function g defined on $\mathbb{R}^*$ by: $g(x) = 3 - \frac{2}{x}$.

1. Complete an array of values for

$$x \in \{-1\,000; -500; -200; -100; -50; 50; 100; 200; 500; 1\,000\}.$$

What can be inferred from this for the behaviour of g in $-\infty$ and in $+\infty$?

2. Complete an array of values for $x = -0{,}5; -0{,}1; -0{,}01; 0{,}01; 0{,}1; 0{,}5$. What can be done about deduce for the behaviour of g 0?
3. Copy and complete the table of values in the manual, then plot the representative curve of function g.

Personal work

Resuming Exercise 3 with the following data:
240 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 25 — Derivative and direction of variation

Reciprocal function

05/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Derivate the reciprocal function.

· Justify variations.

· Draw a picture.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 10 per cent increase .....
2. Derive 3 $x^2 - 5x + 1$. .....
3. Solve $(x-2)(x+1) = 0$. .....
4. To give the prohibited value of $\dfrac{x+1}{x}$. .....
5. Calculate $\dfrac{1}{7}$ and ~~$\dfrac{1}{7}$~~. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: drift the reciprocal function and justify the variations.

Research activity

Based on the rate of increase of $f(x) = \dfrac{1}{x}$, find $f'(x) = -\dfrac{1}{x^2}$. Study his sign and check consistency with the graph.

Trace written

For $x \neq 0$, $f'(x) = -\dfrac{1}{x^2} < 0$. The reciprocal function decreases at each of the intervals $]-\infty, 0[$ and $]0, +\infty[$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{4}{x}$.

Reference in the manual

Supplements: Calao, courses p. 89·93; exercises p. 94, n° 1 to 3, and p. 96, n° 10 to 13. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 94, exercise 3

Calculate the derivatives of the following functions defined on $\mathbb{R}^*$.

a. $f(x) = 2x - 3 + \dfrac{4}{x}$

b. $g(x) = 3x - 5 - \dfrac{1}{x}$

c. $f(x) = 7x^3 - 2x + \dfrac{1}{x}$

d. $g(x) = 3x^2 - 5x - 2 + \dfrac{4}{x}$

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 26 — Combinations with polynomial functions

Reciprocal function

07/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Derive a combination.

· Reduce expression.

· Organize a study.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x+8-x+2$. .....
2. Resolve $(x-2)(x+1)=0$. .....
3. To give the prohibited value of $\dfrac{x+1}{x}$. .....
4. Calculate $\dfrac{1}{8}$ and ~~$\dfrac{1}{8}$~~. .....
5. Derive 3 x^2-5x+1. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: derive a combination and reduce an expression.

Research activity

A company produces $x>0$ hundreds of objects and its average cost is $C(x)=2x+30+\dfrac{200}{x}$. Calculate $C'(x)$, factoring its numerator and seeking the level of production minimizing the cost Medium.

Trace written

To Study $P(x)+\dfrac{k}{x}$, determine the domain, calculate the derivative, reduce to the same denominator, study his sign and then draw up variations.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x)=3+\dfrac{5}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0,+\infty[$ function $g(x)=x+\dfrac{9}{x}$.

Reference in the manual

Supplements: Calao, p. 96, No. 14-21. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 96, exercise 20

Let f the function defined on $\mathbb{R}^*$ by $f(x) = 4x + 1 + \frac{9}{x}$.

1. Show that for all real x Non-zero: $f'(x) = \frac{(2x-3)(2x+3)}{x^2}$.
2. For all real x non-zero, study the sign of $f'(x)$.
3. Deduce changes in function f.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 27 — Signing and factoring

Reciprocal function

12/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Factorize.

· Build a signboard.

· Respect the prohibited values.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 180 .....

2. To give the prohibited value of $\dfrac{x+1}{x}$. .....

3. Calculate $\dfrac{1}{9}$ and ~~$-\dfrac{1}{9}$~~. .....

4. Derive 3 $x^2 - 5x + 1$. .....

5. Solve $(x-2)(x+1) = 0$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: factor and construct a signboard.

Research activity

Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Build the table with the three critical values, distinguish zero and forbidden value, then solve the inequitable $\dfrac{(x-2)(x+3)}{x} \geq 0$.

Trace written

In a quotient, the sign depends on the numerator and denominator.
excluded from the table, even if a factor is formally simplified.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 4 + \dfrac{6}{x}$.

2. Calculate $f'(x)$ and study his sign.

3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{16}{x}$.

Reference in the manual

Supplements: Calao, pp. 96-101. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 96, exercise 21

Let g the function defined on $\mathbb{R}^*$ by $g(x) = x + 10 - \frac{1}{x}$.

1. Show that for all real x Non-zero: $g'(x) = \frac{x^2+1}{x^2}$.
2. For all real x non-zero, study the sign of $g'(x)$.
3. Deduce changes in function g.

Personal work

Resuming Exercise 2 with the following data:
255 and rate or common difference / ratio by 5. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 28 — Derivatives of polynomials of not more than 3 degrees

Reciprocal function

14/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Drain a polynomial.

· Study a derivative.

· Reduce variations.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. Calculate $\frac{1}{3}$ and $-\frac{1}{3}$. . . . . .
3. Derive 3 $x^2 - 5x + 1$. . . . . .
4. Solve $(x-2)(x+1) = 0$. . . . . .
5. To give the prohibited value of $\frac{x+1}{x}$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: drift a polynomial and study a derivative.

Research activity

For $f(x) = 2x^3 - 3x^2 - 12x + 5$, calculate $f'(x)$, factoring, studying his sign and deducing from it the changes in f.

Trace written

$(ax^3 + bx^2 + cx + d)' = 3ax^2 + 2bx + c$. The derivative sign gives the direction of variation of the function.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 5 + \frac{7}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{25}{x}$.

Reference in the manual

Supplements: Calao, pp. 94-96. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 94, exercise 13

Let f the function defined on $\mathbb{R}^*$ by: $f(x) = \frac{1}{x}$.

1. Calculate $f'(x)$.
2. Show that for all real x Non-zero: $f'(x) = \frac{-1}{x^2}$.
3. After studying the sign of $f'(x)$, to draw up the table of variations of f.

Personal work

Resuming Exercise 3 with the following data:
260 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Meeting 29 — Comprehensive study and representation

Reciprocal function

19/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Conduct a comprehensive study.

· Draw a coherent curve.

· Interpret an extreme.

Fluency · 6 to 8 minutes, without calculator

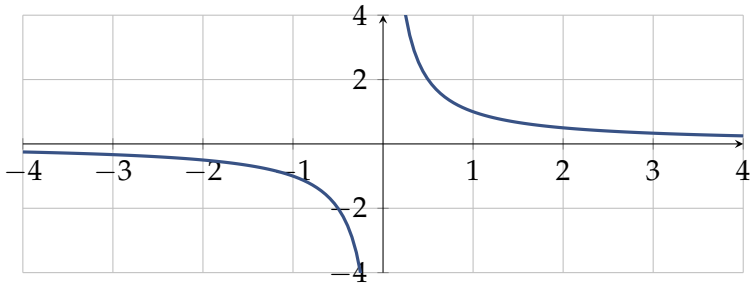
1. Reduce 3 $x+4-x+2$. .....
2. Derive 3 x^2-5x+1. .....
3. Solve $(x-2)(x+1)=0$. .....
4. To give the prohibited value of $\dfrac{x+1}{x}$. .....
5. Calculate $\dfrac{1}{4}$ and ~~$-\dfrac{1}{4}$~~. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: conduct a complete study and plot a curve coherent.

Research activity

Studying completely $g(x)=x+\dfrac{9}{x}$ on $]0,+\infty[$: domain, derivative, sign, variations, minimum, behaviour at the bounds and outline.



Trace written

A complete study includes: domain, derivative, sign, variations, limitations or behaviour in the bounds, remarkable points and consistent graphic.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 6 + \frac{8}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{36}{x}$.

Reference in the manual

Supplements: Calao, pp. 98-103. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 98, financial year 31

A company specialising in industrial ovens can manufacture between 20 and 80 depending on the years. x the number of ovens manufactured annually.
The graphic representation given in the manual is that of a function C defined on L-range $[20;80]$. For all integer values of x, $C(x)$ is the unit cost of production Average (in €).

1. What is the average unit production cost when 25 ovens are produced and then when 70 Ovens are produced?
2. Which productions correspond to an average unit cost of 20,000 € ?
3. What is the minimum unit cost of production? What production does it correspond to?

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Meeting 30 — Assessment and evaluation

Reciprocal function

21/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Mobilize domain, derivative and sign.

· Write a study.

· Correct methodically.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 100 .....
2. Resolve $(x-2)(x+1)=0$. .....
3. To give the prohibited value of $\dfrac{x+1}{x}$. .....
4. Calculate $\dfrac{1}{5}$ and ~~$\dfrac{1}{5}$~~. .....
5. Derive 3 x^2-5x+1. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reformulate: mobilize domain, derivative and sign and write a study.

Research activity

Correcting a deliberately wrong study of $h(x)=x+\dfrac{4}{x}$: area announced $\mathbb{R}$, derivative $1+\dfrac{4}{x^2}$, Tableau sans rupture en 0. Rewrite a complete solution.

Trace written

Control the four objects separately: domain, derivative expression, derivative sign, Variations. A one-step error should not mask the following.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Provide the domain and asymptotes of $f(x)=2+\dfrac{4}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0,+\infty[$ function $g(x)=x+\dfrac{4}{x}$.

Reference in the manual

Supplements: Calao, p. 100 to 103. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 98, exercise 32

Every week, a liquid detergent company estimates that the cost of production (in euros) can be modelled by a function C given by: $C(x) = x^2 + 60x + 121$, where x is the volume of detergent (m) ^{3}), $x \in [1;30]$.
Let f the function representing the average cost of production per m ^{3} liquid detergent.

1. Show that for everything $x \in [1;30]$, $f(x) = x + 60 + \dfrac{121}{x}$.
2. Calculate $f'(x)$. Show that for everything $x \in [1;30]$, $f'(x) = \dfrac{(x-11)(x+11)}{x^2}$.
3. Draw up the table of changes in f.
4. What is the minimum average cost of production? For what amount of detergent is it reached ?

Personal work

Resuming Exercise 3 with the following data:
270 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.